# HAT-P-32b

R-0234

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_251209` · created 2026-07-17T04:05:36+00:00

**BENCHMARK: INACCURATE**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2461018.8289 +/- 0.0036 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.16 +/- 0.019 |
| Transit depth (Rp/Rs)^2 | 2.55 +/- 0.59 [%] |
| Orbital Inclination (inc) | 84.0 +/- 1.73 |
| Airmass coefficient 1 (a1) | 1.253 +/- 0.011 |
| Airmass coefficient 2 (a2) | -0.1046 +/- 0.0073 |
| Scatter in the residuals of the lightcurve fit is | 0.89 % |
| Best Comparison Star | None |
| Optimal Aperture | 3.92 |
| Optimal Annulus | 8.82 |
| Transit Duration (day) | 0.0953 +/- 0.0054 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.16 ± 0.019 vs 0.1489 (deviation 0.58σ) |
| Duration fit vs published | 0.0953 d vs 0.1304 d (deviation 6.45σ) |
| Mid-time O–C | 9.9 min |
| Depth SNR | 8.3 |
| Residual scatter | 0.89 % |

## Light curve

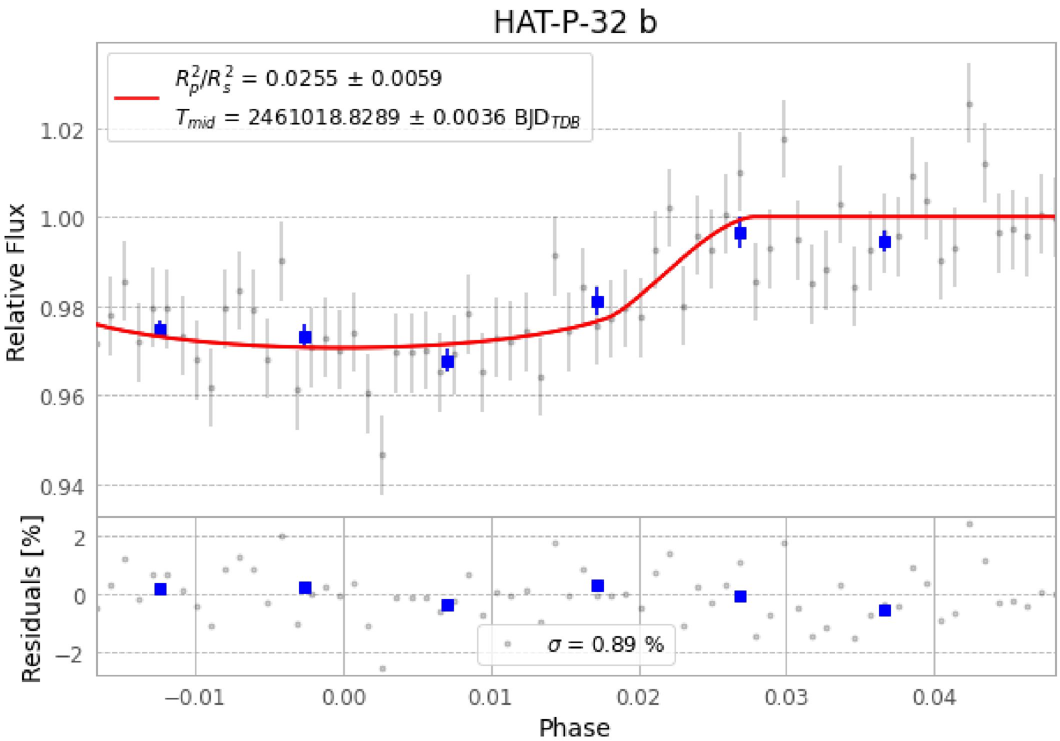


Figure 1 — FinalLightCurve_HAT-P-32 b_2025-12-08.png (SHA-256 85c6b7de1e4f2b13…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [249, 342], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2ba09564a93bdf7c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e`

## Data provenance

68 FITS frames (45 MB), `HATP-32251209065415.FITS` → `HATP-32251209101515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-251209.log (2246eeb83cff…), FinalLightCurve_HAT-P-32 b_2025-12-08.png (85c6b7de1e4f…), FinalLightCurve_HAT-P-32 b_2025-12-08.csv (3416b419022c…)

## Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

  